Problem Set 7

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Problem 7-1.
Subproblems: $x(i)$: max delegates Torr can win in states i to n, where campaigning on day i is possible, for $i \in \{1, \ldots, n+1\}$.

Relate: $x(i) = \max\{\, d_i + z_{i+1} + z_{i+2} + x(i+3),\ \ z_i + x(i+1) \,\}$

Topological Order: $n+1, n, \ldots, 2, 1$ ($x(i)$ only depends on subproblems with strictly larger i)

Base Cases: $x(n+1) = 0, x(n) = d_n, x(n-1) = \max\{\, d_{n-1} + z_n,\ z_{n-1} + d_n \,\}$

Original Problem: $x(1) \geq \lfloor D/2 \rfloor + 1$

Time: $T(n) = T(n-1) + O(1) \implies O(n)$

Problem 7-2.
Subproblems: $x(i, j)$: min total discomfort matching tigers $T[i:]$ to a subsequence of $C[j:]$, for $i \in \{0, \ldots, n\}$, $j \in \{0, \ldots, n^2\}$

Relate: $x(i,j) = \min\{\, d(i,j) + x(i+1, j+1),\ \ x(i, j+1) \,\}$, where $d(i,j) = \max\{0,\ s_i - c_j\}$

Topological Order: $j = n^2, n^2 - 1, \ldots, 1, 0$ ($x(i,j)$ only depends on subproblems with strictly larger j)

Base Cases: $x(n, j) = 0,\ \ x(i, n^2) = \infty$ for $i > 0$

Original Problem: $x(0, 0)$

Time: : (n+1)(n²+1) subproblems $\times\ O(1) \implies O(n^3)$

Problem 7-3.
Subproblems: $x(v, i)$: number of paths from s to v with even weight if $i = 0$, odd weight if $i = 1$, for $v \in V$, $i \in \{0, 1\}$.

Relate: $x(v,i) = \displaystyle\sum_{u \,\in\, \text{Adj}^-(v)} x\big(u,\ p(w(u,v) + i)\big)$, where $p(k)$ is the parity of k

Topological Order: topological order of G ($x(v,i)$ only depends on subproblems for vertices earlier in the order)

Base Cases: $x(s, 0) = 1,\ \ x(s, 1) = 0,\ \ x(v, 0) = x(v, 1) = 0$ for v with $\text{Adj}^-(v) = \emptyset$

Original Problem: $x(t, 1)$

Time: $\displaystyle\sum_{v \in V} O(\deg^-(v)) = O(|V| + |E|)$

Problem 7-4.

Subproblems: $x(i, j, p)$: max tastiness Liza can achieve with j slices counter-clockwise from α_i remaining, Liza choosing if $p = 1$, Lie choosing if $p = 2$, for $i \in \{0, \ldots, 2n-1\}$, $j \in \{0, \ldots, n\}$, $p \in \{1, 2\}$

Relate:

$$x(i,j,1) = \max_{k \in \{1,\ldots,j-1\}} \big\{\, v(i,k) + x(i+k,\ j-k,\ 2),\ \ v(i+k,\ j-k) + x(i,\ k,\ 2) \,\big\}$$

$$x(i,j,2) = \min_{k \in \{1,\ldots,j-1\}} \big\{\, x(i+k,\ j-k,\ 1),\ \ x(i,\ k,\ 1) \,\big\}$$

where $v(i,j) = \sum_{k=0}^{j-1} t_{(i+k) \bmod 2n}$ (precomputed for all i, j in $O(n^3)$ time)

Topological Order: $j = 1, 2, \ldots, n$ ($x(i, j, p)$ only depends on subproblems with strictly smaller j)

Base Cases: $x(i, 1, 1) = t_i$, $\ x(i, 1, 2) = 0$

Original Problem: $\max_{i \in \{0,\ldots,2n-1\}} \big\{\, x(i, n, 2) + v\big((i+n) \bmod 2n,\ n\big) \,\big\}$

Time: $O(n^3)$ to compute all $v(i, j)$, plus $O(n^2)$ subproblems $\times\ O(n) \implies O(n^3)$

Problem 7-5.

(a)

Subproblems: $x(j)$: length of longest decreasing no-skip-days subsequence in $P[j:]$ that includes $P[j]$, for $j \in \{0, \ldots, nk-1\}$

Relate: $x(j) = 1 + \max\big(\{x(d) \mid d \in \{j+1, \ldots, f(j)\},\ P[j] > P[d]\} \cup \{0\}\big)$, where $f(j) = \min\{j + (k-1) - (j \bmod k) + k,\ nk - 1\}$

Topological Order: $j = nk-1, nk-2, \ldots, 1, 0$ ($x(j)$ only depends on subproblems with strictly larger j)

Base Cases: $x(nk-1) = 1$ (also implied by relation when the max is over the empty set)

Original Problem: $\max_{j=0}^{nk-1} x(j)$, over all companies; store parent pointers to reconstruct S

Time: $O(nk)$ subproblems $\times\ O(k) \implies O(nk^2)$ per company, $O(snk^2)$ total